Computer Science and Engineering
CSE221 Database Systems
Dr. Laila Shoukry

Project Topic and Deadlines Spring 2024

In this group project you will gain experience in the process of relational database design and development. It is worth 25% of the course grade. There can be up to 5 students per team. Each student should be responsible for one or more phases of the group, according to number of students per team.

The following will be the main phases of the project, deliverable deadlines and their respective tentative weights:
Phase 1: Business Rules and Requirements: (4 pts) 21st of March
Phase 2: EERD: (4 pts) 30th March
Phase 3: Normalized Relational Schema: (4 pts) 6th of April
Phase 4: Implementation DDL DML: (4 pts) 15th of April
Phase 5: Views, Forms, Reports, Sheet Upload and Export: (4 pts) 22th of April
Final Report and Discussion: (5 pts) 29th of April
(Extracurricular): Fully Connected Website, Video – Check Rubric (10 pts) 6th of May

For the project phases descriptions and rubrics, please check the project file.
You need to choose your teams until 16th of March, otherwise you will get deducted from the marks of Phase 1!

Project Topic Description Spring 2024

In this project you will create and query a database for AIU Coursera Courses.
This is the description of the course and you may need further information which you can get from the example data and reports provided:
Every semester (Fall/Spring/Summer +Year) we need to keep track of AIU courses, their field (CSE, Pharmacy, Dentistry, Business, Public Health, Art and Design), their course code, their semester in the study plan, their main instructor this semester and his chosen Coursera course(s) for the course and the number of expected students in his AIU course (so also in Coursera) this semester.
Each instructor can choose more than one Coursera course for each of his AIU courses.

For each Coursera course (also each guided project), we need to store its name, its instructor name, its link (the last part of the link is unique, it is called the course slug), the organization or university which offers this course, its number of weeks, number of hours, and, if it is part of a specialization (only for courses, not for guided projects), the link of the specialization.
We need to know if a certain AIU course used Coursera before in one of the previous semesters, and if yes, which course(s) were used (with all their info).

Every AIU field has a program on Coursera where the students can be invited according to their field (CSE, Pharmacy, ...all AIU Fields) and each of these programs has an admin link for the instructors (who are invited as admins) and a learner link for students (who are invited as learners). We collect a list of all instructors from each field every semester to add them as admins on their programs and also as admins and learners on the Instructors program (another Coursera program created for instructors).

The instructor can choose certain modules from Coursera and map them to certain weeks of his course. We need to store this mapping into our catalog and also calculate the overlap percentage from both sides: how much is covered from the AIU course and how much from the Coursera course. If it is a guided project, it can be added to one or two weeks in the AIU course schedule.

Every start of a semester, we need to have a report about each AIU field: How many AIU courses have requested Coursera this semester, what are the course codes, how many students for each course, how many students for each level, the duration (start and end time) when they need the Coursera licenses, then calculate how many students will use Coursera overall during every of the 16 semester weeks of the AIU course.

Then every week we should be able to see which students need to be added now to add them if there are enough Coursera licenses remaining (AIU has a certain quota for the number of students who can use it at the same time). The student lists registered in an AIU course are provided by each instructor to invite them, the list contains email, name and ID. These can be provided as links to csv files uploaded on forms to Onedrive. Also, every week we need to check which students finished their duration and should be removed. Also, from the Coursera sheet we can get information about how many students joined the program from the invitation, how many enrolled in each Coursera course and how many completed each course and update this info weekly in the database reports.

The weekly report needs to have the total of students invited until now from each field and how many of them were removed, also the total from all fields together and the total number of enrolled students and completions for each field and altogether. Then at the end of the semester we need the total of the semester of these numbers per field + the total for AIU. We also need to know students who have not enrolled in any course in Coursera to remove them.

At the end of the year, we need to know the students who had the highest number of certificates from each field and each level.
If these reports are not generated automatically, we need a way at least to query the database for this info afterwards.
The system can be used by the main Coursera coordinator (can see everything), the coordinators from the different fields (can see their fields only) and the instructors (can see their courses only and Coursera courses which were used before in their courses).